Aerial Survey ROI

Technical Guide

Transect Planning Workflow — Configuration & Reference

Version 1.1 · Generated July 21, 2026

1. Overview

The **Aerial Survey ROI** workflow generates parallel transect lines across a user-defined polygon boundary at a configurable compass bearing and spacing. Outputs are exported as geospatial files and rendered as an interactive map widget in the Ecoscope dashboard.

- **survey_lines_<hash>.parquet** — transect lines in GeoParquet format
- **aerial_survey.html** — interactive HTML ecomap
- **Dashboard widget** — map titled *Aerial Survey Lines*

2. Prerequisites

2.1 Packages

As of the migration to the **ecoscope-platform** task/compiler runtime, the workflow's requirements are:

Package	Version	Channel
ecoscope-platform	>=2.15.0, <2.16.0	https://repo.prefix.dev/ecoscope-workflows/
ecoscope-workflows-ext-custom	0.1.0rc14.*	https://repo.prefix.dev/ecoscope-workflows-custom/
ecoscope-workflows-ext-ste	0.0.0rc1.*	https://repo.prefix.dev/ecoscope-workflows-custom/
pydeck	0.9.2	conda-forge

Note: ecoscope-workflows-core and ecoscope-workflows-ext-ecoscope are no longer direct dependencies — their task coverage now lives in ecoscope-platform. Unlike some other STE workflows, this one does not depend on opentelemetry-sdk.

2.2 Environment

Variable	Required	Description
ECOSCOPE_WORKFLOWS_RESULTS	Yes	Writable directory for all output files.

2.3 Input File

Format	Geometry	CRS
GeoPackage (.gpkg), GeoJSON (.geojson), GeoParquet	Polygon or MultiPolygon	Any EPSG — reprojected to EPSG:4326 automatically.

Note: Files with no CRS assigned will raise a projection error. Set an EPSG code before uploading.

3. Configurable Parameters

Four task groups are user-configurable. All remaining tasks run with fixed defaults.

3.1 Set Workflow Details

Task ID: workflow_details · **Function:** set_workflow_details

Name and description for this run. Appears in the dashboard header.

Field	Type	Example
Name	string	Mara North Conservancy Aerial Survey
Description	string	Pre-season transect planning, April 2026

3.2 Time Range (optional)

Task ID: time_range · **Function:** set_time_range

Does **not** filter the ROI or transect geometry. Recorded in the dashboard for traceability against a specific survey window.

Field	Format	Example
Since	ISO 8601	2026-02-03T10:14:00.000Z
Until	ISO 8601	2026-03-03T10:14:00.000Z
Timezone	IANA label	UTC
Time format	strftime string	%d %b %Y %H:%M:%S

3.3 Define Region of Interest

Task group · **Task ID:** retrieve_file_params · **Function:** get_file_path

Provide the boundary file as a URL or local path. The file is loaded into a GeoDataFrame and styled as a map layer.

Input method	Example
URL	https://example.com/my_conservancy.gpkg
Local path	/data/inputs/my_conservancy.gpkg

Automatic tasks after load

Task ID	Function	Action
load_gdf	load_df	Load file into GeoDataFrame.
generate_layers_map	create_geojson_layer	Create ROI boundary layer — olive green (#556b2f), 25 % opacity.

3.4 Draw Aerial Survey Lines

Task group · **Task ID:** survey_lines · **Function:** ecoscope_workflows_ext_ste.tasks.spatial_operations.create_survey_transects

Generates parallel transect lines tiled across the ROI bounding box at the given bearing, then clips them to the ROI polygon. Lines outside the boundary are discarded.

Parameter	UI Label	Type	Description	Example
direction	Survey Direction (degrees)	compass choice	Compass bearing the lines will run. 0° = N–S, 90° = E–W.	North–South
spacing	Line Spacing (m)	number	Distance in metres between adjacent transect lines.	1500

Spacing guidance

ROI size	Suggested spacing
Small (< 100 km²)	500 – 1 000 m
Medium (100–1 000 km²)	1 000 – 2 500 m
Large (> 1 000 km²)	2 500 – 5 000 m

Note: If spacing exceeds the cross-track ROI extent, create_survey_transects returns an empty GeoDataFrame and all downstream tasks are skipped. Reduce spacing and re-run.

Line style

Property	Value
Colour	Yellow #FFFF00 RGB(255, 255, 0)
Opacity	55 %
Width	1–5 px (pixel-scaled)
Display CRS	EPSG:4326

3.5 Configure Base Map Layers

Task ID: configure_base_maps · **Function:** set_base_maps_pydeck

Two stacked ArcGIS tile layers form the background of the ecomap. Pre-filled with defaults, but user-editable.

Layer	Default Opacity	Max Zoom
ESRI World Hillshade	1.0	20
ESRI World Street Map	0.15	20

4. Workflow DAG

Tasks are chained via ${{ workflow.<id>.return }} references. The table lists every node, its upstream dependencies, and its role.

Task ID	Depends on	Role
workflow_details	—	Run name / description
time_range	—	Reporting time window
groupers	—	Attribute groupers (fixed: empty)
configure_base_maps	—	ArcGIS tile layer configuration
retrieve_file_params	—	Resolve ROI file URL or path
load_gdf	retrieve_file_params	Load ROI into GeoDataFrame
generate_layers_map	load_gdf	ROI boundary DeckGL layer (olive green)
survey_lines	load_gdf	Generate transect lines
filter_columns	survey_lines	Drops fid/FID columns (defined but currently unused downstream
persist_aerial_gpq	survey_lines	Write survey_lines_<hash>.parquet

Task ID	Depends on	Role
transform_gdf	survey_lines	Reproject lines to EPSG:4326
transform_load_gdf	load_gdf	Reproject ROI to EPSG:4326
aerial_survey_polylines	transform_gdf	Survey lines DeckGL layer (yellow)
zoom_gdf_extent	transform_load_gdf	Fit viewport to ROI extent
draw_aerial_survey_lines_ecomap	configure_base_maps, generate_layers_map, aerial_survey_polylines, zoom_gdf_extent	Render HTML ecomap
persist_ecomaps	draw_aerial_survey_lines_ecomap	Write aerial_survey.html
create_aerial_widgets	persist_ecomaps	Wrap map in dashboard widget
patrol_dashboard	workflow_details, create_aerial_widgets, time_range, groupers	Assemble dashboard

Note: filter_columns is defined in spec.yaml but its output is not currently wired into any downstream task — both persist_aerial_gpq and transform_gdf read from survey_lines directly, so the exported columns are not filtered.

5. Outputs

All files are written to $ECOSCOPE_WORKFLOWS_RESULTS.

File	Format	Use
survey_lines_<hash>.parquet	GeoParquet	Python / GeoPandas, cloud storage
aerial_survey.html	HTML	Browser — interactive PyDeck ecomap
Dashboard widget	Ecoscope	In-app reporting

A content hash is appended to the GeoParquet filename to avoid overwriting previous runs' output.

Map layer styles

Layer	Colour	Opacity	Width
ROI boundary	Olive green #556b2f	25 % fill	1.00 px
Survey transects	Yellow #FFFF00	55 %	1–5 px
World Hillshade	ArcGIS tile	100 %	—
World Street Map	ArcGIS tile	15 %	—

6. Skip Logic

All tasks share a global skipif policy (task-instance-defaults in spec.yaml):

```
skipif:
  conditions:
    - any_is_empty_df
    - any_dependency_skipped
```

Condition	Trigger	Effect
any_is_empty_df	Any upstream GeoDataFrame is empty.	Task and all dependants are skipped, not crashed.

Condition	Trigger	Effect
any_dependency_skipped	Any upstream task was skipped.	Skip propagates transitively through the DAG.

Note: Skipped tasks show in the runner UI with a skipped status — not an error. Confirm all tasks completed to verify a full successful run.

7. Troubleshooting

Symptom	Cause	Fix
Tasks skipped downstream	Empty GeoDataFrame from ROI load or create_survey_transects	Check file_path/URL; verify polygon features in QGIS; reduce spacing.
No lines on map	Spacing too large for the ROI extent.	Halve the spacing value and re-run.
CRS error	Source file has no CRS.	QGIS: Layer Properties → Set CRS. Python: gdf.set_crs('EPSG:4326').
File not found	URL inaccessible or requires authentication.	Use a direct download link (e.g. Dropbox ?dl=1).
Missing output files	ECOSCOPE_WORKFLOWS_RESULTS unset or not writable.	Export the variable; confirm directory write permissions.
Blank HTML map	Browser CORS block on local tile requests.	Serve with: python -m http.server 8080

8. Example Configuration

From test-cases.yaml — scenario: *all-grouper* (Mara North Conservancy):

```
workflow_details:
  name: Custom

time_range:
  timezone: { label: UTC, tzCode: UTC, utc: '+00:00' }
  time_format: '%d %b %Y %H:%M:%S'
  since: '2026-02-03T10:14:00.000Z'
  until: '2026-02-03T10:14:00.000Z'

retrieve_file_params:
  input_method:
    url: 'https://www.dropbox.com/.../mnc_conservancy.gpkg'

survey_lines:
  direction: 'North South'
  spacing: 1500
```

- N–S transect lines, 1 500 m apart, clipped to the MNC boundary.
- Outputs: survey_lines_<hash>.parquet, aerial_survey.html.
- Dashboard widget titled *Aerial Survey Lines*.

9. Software Versions

Package	Version	Role
ecoscope-platform	>=2.15.0, <2.16.0	Workflow engine and core task library

Package	Version	Role
ecoscope-workflows-ext-custom	0.1.0rc14.*	Custom tasks including create_geojson_layer
ecoscope-workflows-ext-ste	0.0.0rc1.*	STE-specific tasks including create_survey_transects
pydeck	0.9.2	Renders the interactive DeckGL ecomap

ecoscope-platform is pulled from the ecoscope-workflows prefix.dev channel; the two ecoscope-workflows-ext-* packages from ecoscope-workflows-custom; pydeck from conda-forge. Packages are pinned to compatible versions and managed by **pixi**.